



Compte Rendu

Tp 5

La généricité& la gestion des exceptions

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Partie 1:

Exercice 1 (*Division sécurisée*) :

```
package partiel1;

import java.util.Scanner;

class Calcule {

    public int division(int a, int b) {
        if (b == 0) {
            throw new ArithmeticException("Division par zéro interdite ");
        }

        return a / b;
    }
}

public class ex1 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        try {

            System.out.print("entrez le premier nombre : ");
            int x = sc.nextInt();

            System.out.print("entrez le deuxième nombre : ");
            int y = sc.nextInt();

            Calcule c = new Calcule();

            int resultat = c.division(x, y);

            System.out.println("resultat = " + resultat);

            catch (ArithmeticException e) {
                System.out.println("Erreur : " + e.getMessage());
            }

            catch (Exception e) {
                System.out.println("nombre invalid");
            }

        }
    }
}
```

```
entrez le premier nombre : 4
entrez le deuxième nombre : 2
resultat = 2
```

Exercice2:

```
package partie1;

class text{
    public int longueur(String chaine) {
        if(chaine == null) {
            System.out.println("la chaine est null");
            return 0;
        }
        return chaine.length();
    }
}

public class ex2 {
    public static void main(String [] args ) {

        text t = new text ();

        String mot1 = "hiba";

        int resultat1 = t.longueur(mot1);
        System.out.println("Longueur = " + resultat1);

        String mot2 = null;

        int resultat2 = t.longueur(mot2);

        System.out.println("Longueur = " + resultat2);

    }
}
```

```

package partie1;

class texte {
    public int longueur(String chaine) {
        try {
            return chaine.length();
        }
        catch (NullPointerException e) {
            System.out.println("la chaine est null");
            return 0;
        }
    }
}

public class ex2v2 {
    public static void main(String[] args) {
        text t = new text();

        String mot1 = "jebabra";

        int resultat1 = t.longueur(mot1);

        System.out.println("longueur = "+resultat1);

        String mot2 = null;

        int resultat2 = t.longueur(mot2);

        System.out.println("longueur =" +resultat2);

    }
}

```

```

Longueur = 4
la chaine est null
Longueur = 0

```

Exercice3:

```

package partie1;
import java.util.Scanner;

public class ex3 {

    public static void accesIf(int[] tab, int index) {
        if (index >= 0 && index < tab.length) {
            System.out.println(tab[index]);
        } else {
            System.out.println("erreur");
        }
    }

    public static void accesTry(int[] tab, int index) {
        try {
            System.out.println(tab[index]);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Index hors limites");
        }
    }

    public static void main(String[] args) {
        int[] tab = {4,2,3,4,5};
        accesIf(tab, 5);
        accesTry(tab, 25);
    }
}

```

```

erreur
Index hors limites

```

Exercice4:

```

1 package partie1;
2
3 public class ex4 {
4
5     public static int convertir(String ch) {
6
7         try {
8             return Integer.parseInt(ch);
9
10        } catch (NumberFormatException e) {
11            System.out.println("conversion impossible");
12            return 0;
13        }
14    }
15
16    public static void main(String[] args) {
17
18        int x = convertir("51");
19
20        System.out.println(x);
21
22        convertir("nom");
23    }
24 }

```

terminated / ex4 [Java Application] C:\Users\j\Documents\p2\p001\plugins\org.eclipse.jdt.ui.openjdk\bin\spotbugs.jar

51
conversion impossible

Exercice5:

```

1 package partie1;
2
3 public class ex5 {
4
5     public static int racineCarree(int x) {
6
7         if (x < 0) {
8             throw new IllegalArgumentException("nombre negatif");
9         }
10
11        return (int) Math.sqrt(x);
12    }
13
14    public static void main(String[] args) {
15
16        try {
17            System.out.println(racineCarree(49));
18
19            System.out.println(racineCarree(-9));
20
21        } catch (IllegalArgumentException e) {
22            System.out.println(e.getMessage());
23        }
24    }
25 }

```

7
nombre negatif

Exercice6:

```
1 package partie1;
2
3 class machine6{
4     private boolean etat = false;
5
6     public void demarrer() {
7
8         if (etat) {
9             throw new IllegalStateException("machine ON");
10        }
11
12        etat = true;
13
14        System.out.println("machine démarrée");
15    }
16 }
17
18 public class ex6 {
19
20     public static void main(String[] args) {
21
22         machine6 m = new machine6();
23
24         try {
25             m.demarrer();
26             m.demarrer();
27         } catch (IllegalStateException e) {
28             System.out.println(e.getMessage());
29         }
30     }
31 }
```

machine démarrée
machine ON

<terminated> ex6 [Java Application] C:\Users\HIBA\p2\p...

Exercice 7:

```

1 package partie1;
2
3 public class ex7 {
4
5     public static void C() {
6         throw new ArithmeticException("erreur en C");
7     }
8
9     public static void B() {
10        C();
11    }
12
13    public static void A() {
14        try {
15            B();
16        } catch (ArithmeticException e) {
17            System.out.println("exception en A");
18        }
19    }
20
21    public static void main(String[] args) {
22        A();
23    }
24 }

```

exception en A

Exercice 8:

```

package partie1;

public class ex8 {

    public static void verifierAge(int age) throws Exception {
        if (age < 18) {
            throw new Exception("age inférieur a 18");
        }

        System.out.println("age valid");
    }

    public static void main(String[] args) {
        try {
            verifierAge(10);
        } catch (Exception e) {
            System.out.println(e.getMessage());
        }
    }
}

```


Partie 2:

Exercice 1:

```
3 class SoldeInsuffisantException extends Exception {
4
5     public SoldeInsuffisantException(String msg) {
6         super(msg);
7     }
8 }
9
10 class CompteBancaire {
11     private double solde;
12
13     public CompteBancaire(double solde) {
14         this.solde = solde;
15     }
16
17     public void retirer(double montant)
18         throws SoldeInsuffisantException {
19
20         if (montant > solde) {
21             throw new SoldeInsuffisantException("Solde insuffisant");
22         }
23
24         solde -= montant;
25     }
26 }
27
28 public class ex1 {
29
30     public static void main(String[] args) {
31
32         CompteBancaire c = new CompteBancaire(1000);
33
34         try {
35
36             c.retirer(2600);
37
38         } catch (SoldeInsuffisantException e) {
39
40             System.out.println(e.getMessage());
41         }
42     }
43 }
```

solde insuffisant

Exercice2:

```

package partie2;

class MontantInvalideException extends Exception {

    public MontantInvalideException(String msg) {
        super(msg);
    }
}

```

Exercice3:

```

package partie2;

class MontantInvalideException extends Exception {
    public MontantInvalideException(String msg) {
        super(msg);
    }
}

class SoldeInsuffisantException extends Exception {
    public SoldeInsuffisantException(String msg) {
        super(msg);
    }
}

class Compte {
    double solde = 1000;

    public void retirer(double montant)
        throws MontantInvalideException,
               SoldeInsuffisantException {

        if (montant <= 0) {
            throw new MontantInvalideException("montant invalide");
        }

        if (montant > solde) {
            throw new SoldeInsuffisantException("solde insuffisant");
        }

        solde -= montant;
    }
}

public class ex3{

    public static void main(String[] args) {

        Compte c = new Compte();

        try {
            c.retirer(-9);
        } catch (Exception e) {
            System.out.println(e.getMessage());
        }
    }
}

```

```
montant invalide
```

Exercise 4:

```
package partiez;

class EmailInvalideException extends Exception {
    public EmailInvalideException(String msg) {
        super(msg);
    }
}

class AgeInvalideException extends Exception {
    public AgeInvalideException(String msg) {
        super(msg);
    }
}

public class ex4 {

    public static void inscrire(String email, int age)
        throws EmailInvalideException,
        AgeInvalideException {

        if (!email.contains("@")) {
            throw new EmailInvalideException("email invalide");
        }

        if (age < 18) {
            throw new AgeInvalideException("age invalide");
        }

        System.out.println("Inscription reussie");
    }

    public static void main(String[] args) {

        try {
            inscrire("hiba", 20);
        } catch (Exception e) {

            System.out.println(e.getMessage());
        }
    }
}
```

```
email invalide
```

Exercise 5:

```

1 package partie2;
2
3 class AuthentificationException extends Exception {
4
5     public AuthentificationException(String msg) {
6         super(msg);
7     }
8 }
9
10 public class ex5 {
11
12     public static void login(String user, String pass)
13         throws AuthentificationException {
14
15         if (!user.equals("hiba")
16             || !pass.equals("1234")) {
17
18             throw new AuthentificationException(
19                 "identifiants incorrects");
20         }
21
22         System.out.println("connexion réussie");
23     }
24
25     public static void main(String[] args) {
26
27         try {
28
29             login("log", "2020");
30
31         } catch (AuthentificationException e) {
32
33             System.out.println(e.getMessage());
34         }
35     }
36 }

```

identifiants incorrects

Exercice6:

```

1 package partie2;
2
3 class StockInsuffisantException extends Exception {
4
5     public StockInsuffisantException(String msg) {
6         super(msg);
7     }
8 }
9
10 class Produit {
11
12     int stock = 3;
13
14     public void retirerDuStock(int qte)
15         throws StockInsuffisantException {
16
17         if (qte > stock) {
18             throw new StockInsuffisantException(
19                 "stock insuffisant");
20         }
21
22         stock -= qte;
23     }
24 }
25
26 public class ex6 {
27
28     public static void main(String[] args) {
29
30         Produit p = new Produit();
31
32         try {
33
34             p.retirerDuStock(12);
35
36         } catch (Exception e) {
37
38             System.out.println(e.getMessage());
39         }
40     }
41 }

```

stock insuffisant

Exercice 7:

```
1 package partie2;
2
3 class QuotaDepasseException extends Exception {
4
5     public QuotaDepasseException(String msg) {
6         super(msg);
7     }
8 }
9
10 public class ex7 {
11
12     public static void telechargerFichier(double taille)
13         throws QuotaDepasseException {
14
15         double limite = 200;
16
17         if (taille > limite) {
18             throw new QuotaDepasseException(
19                 "quota depasee");
20         }
21
22         System.out.println("telechargement OK");
23     }
24
25     public static void main(String[] args) {
26
27         try {
28
29             telechargerFichier(380);
30
31         } catch (Exception e) {
32
33             System.out.println(e.getMessage());
34         }
35     }
36 }
```

quota depasee

Exercice 8:

```

package partie2;

class ChampObligatoireException extends Exception {

    public ChampObligatoireException(String msg) {
        super(msg);
    }
}

public class ex8 {

    public static void validerFormulaire(
        String nom,
        String email)
        throws ChampObligatoireException {

        if (nom.isEmpty() || email.isEmpty()) {

            throw new ChampObligatoireException(
                "remplir la case il est vide");
        }

        System.out.println("formulaire rempli");
    }

    public static void main(String[] args) {

        try {

            validerFormulaire("", "a@gmail.com");

        } catch (Exception e) {

            System.out.println(e.getMessage());
        }
    }
}

```

remplir la case il est vide

Exercice 9:

```

1 package partie2;
2
3 class PaiementRefuseException extends Exception {
4
5     public PaiementRefuseException(String msg) {
6         super(msg);
7     }
8 }
9
10 class CarteExpireeException extends Exception {
11
12     public CarteExpireeException(String msg) {
13         super(msg);
14     }
15 }
16
17 public class ex9 {
18
19     public static void payer(
20         double montant,
21         boolean expiree)
22         throws PaiementRefuseException,
23             CarteExpireeException {
24
25         if (montant > 5000) {
26             throw new PaiementRefuseException(
27                 "montant dépasse les limite");
28         }
29
30         if (expiree) {
31             throw new CarteExpireeException(
32                 "carte expiree");
33         }
34
35         System.out.println("paiement valid");
36     }
37
38     public static void main(String[] args) {
39
40         try {
41             payer(7000, true);
42         } catch (Exception e) {
43             System.out.println(e.getMessage());
44         }
45     }
46 }

```

montant dépasse les limite

Exercice 10:

Exercice 10 — Conception

Quand utiliser checked ?

Quand l'erreur doit obligatoirement être gérée.

Exemple :

- Lecture fichier
- Base de données

Quand utiliser unchecked ?

Quand l'erreur vient d'une mauvaise utilisation du programme.

Exemple :

- division par zéro
- index invalide

Pourquoi créer une exception personnalisée ?

Pour :

- rendre le code plus clair
- identifier précisément l'erreur
- faciliter la maintenance

partie 3:

exercice1:


```

1 package partie3;
2
3 class Boite<T> {
4
5     private T contenu;
6
7     public void setContenu(T contenu) {
8         this.contenu = contenu;
9     }
10
11     public T getContenu() {
12         return contenu;
13     }
14 }
15
16 public class ex1 {
17
18     public static void main(String[] args) {
19
20         Boite<String> b1 = new Boite<>();
21
22         b1.setContenu("java");
23
24         System.out.println(b1.getContenu());
25
26         Boite<Integer> b2 = new Boite<>();
27
28         b2.setContenu(18);
29
30         System.out.println(b2.getContenu());
31     }
32 }

```

```

java
18

```

Exercice 2:

```

1 package partie3;
2
3 class Paire<T,U> {
4
5     T premier;
6     U second;
7
8     public Paire(T premier, U second) {
9         this.premier = premier;
0         this.second = second;
1     }
2
3     public void afficherPaire() {
4
5         System.out.println(
6             premier + " - " + second);
7     }
8 }
9
0 public class ex2{
1
2     public static void main(String[] args) {
3
4         Paire<String,Integer> p =
5             new Paire<>("hiba", 15);
6
7         p.afficherPaire();
8     }
9 }

```

hiba - 15

Exercice 3:

```

1 package partie3;
2
3 interface Calcul<T> {
4
5     T addition(T a, T b);
6 }
7
8 class CalculInteger implements Calcul<Integer> {
9
10     public Integer addition(Integer a, Integer b) {
11
12         return a + b;
13     }
14 }
15
16 class CalculDouble implements Calcul<Double> {
17
18     public Double addition(Double a, Double b) {
19
20         return a + b;
21     }
22 }
23
24 public class ex3{
25
26     public static void main(String[] args) {
27
28         CalculInteger c1 = new CalculInteger();
29
30         System.out.println(c1.addition(8,3));
31
32         CalculDouble c2 = new CalculDouble();
33
34         System.out.println(c2.addition(4.5,1.7));
35     }
36 }

```

11

6.2

Exercise 4:

```

1 package partie3;
2
3 interface Compareur<T> {
4     int comparer(T a, T b);
5 }
6
7 class CompareurInteger
8     implements Compareur<Integer> {
9
10     public int comparer(Integer a, Integer b) {
11
12         return a.compareTo(b);
13     }
14 }
15
16 class CompareurString
17     implements Compareur<String> {
18
19     public int comparer(String a, String b) {
20
21         return Integer.compare(
22             a.length(),
23             b.length());
24     }
25 }
26
27 public class ex4 {
28     public static void main(String[] args) {
29
30         CompareurInteger c1 =
31             new CompareurInteger();
32
33         System.out.println(c1.comparer(8,2));
34
35         CompareurString c2 =
36             new CompareurString();
37
38         System.out.println(
39             c2.comparer("hiba","java"));
40     }
41 }

```

1
0

Exercice 6:

```

package partie3;

public class ex6 {

    public static <T> void afficherTableau(T[] tableau) {
        for (T x : tableau) {
            System.out.println(x);
        }
    }

    public static <T> T getPremier(T[] tableau) {
        return tableau[0];
    }

    public static void main(String[] args) {
        Integer[] tab = {4, 2, 9};
        afficherTableau(tab);
        System.out.println(getPremier(tab));
    }
}

```

```

4
2
9
4

```

Exercise 7:

```

package partie3;

public class ex7 {

    public static <T extends Number>
    double somme(T a, T b) {
        return a.doubleValue()
            + b.doubleValue();
    }

    public static void main(String[] args) {
        System.out.println(
            somme(9, 3));

        System.out.println(
            somme(3.5, 7.6));
    }
}

```

```

12.0
11.1

```

Exercise 8:

```

package partie3;

class Animal<T> {
    T nom;
}

class Chien extends Animal<String> {
}

public class ex8 {
    public static void main(String[] args) {
        Chien c = new Chien();
        c.nom = "alex";
        System.out.println(c.nom);
    }
}

```

alex

Exercice 9:

```

package partie3;

class Vehicule<T> {
    T vitesse;
}

class Voiture<T>
    extends Vehicule<T> {
}

public class ex9 {
    public static void main(String[] args) {
        Voiture<Integer> v =
            new Voiture<>();
        v.vitesse = 80;
        System.out.println(v.vitesse);
    }
}

```

80

Exercice 10:

```

3 class User {
4     String nom;
5 }
6
7 class Repository<T> {
8
9     void save(T obj) {
10
11         System.out.println("objet enregisterer");
12     }
13 }
14
15 class UserRepository
16     extends Repository<User> {
17
18 }
19
20 public class ex10 {
21
22     public static void main(String[] args) {
23
24         UserRepository ur =
25             new UserRepository();
26
27         ur.save(new User());
28     }
29 }

```

objet enregisterer

Exercise 11:

```

1 package partie3;
2
3 import java.util.List;
4
5 public class ex11 {
6
7     public static void afficherListe(
8         List<?> liste) {
9
10         for (Object x : liste) {
11
12             System.out.println(x);
13         }
14     }
15
16     public static void main(String[] args) {
17
18         afficherListe(
19             Arrays.asList(5,7,8));
20
21         afficherListe(
22             Arrays.asList("ah","no"));
23     }
24 }

```

5
7
8
ah
no

Exercise 12:

```
1 package parties;  
2  
3 import java.util.List;  
4  
5  
6 public class ex12 {  
7  
8     public static void afficherNombres(  
9         List<? extends Number> liste) {  
10  
11         for (Number x : liste) {  
12             System.out.println(x);  
13         }  
14     }  
15  
16  
17     public static void main(String[] args) {  
18  
19         List<Integer> l1 =  
20             Arrays.asList(1,2,3);  
21  
22         List<Double> l2 =  
23             Arrays.asList(1.1,2.2);  
24  
25         afficherNombres(l1);  
26  
27         afficherNombres(l2);  
28     }  
29 }
```

1
2
3
1.1
2.2